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Evaluation of image processing technique as an expert system in mulberry fruit grading based on ripeness level using artificial neural networks (ANNs) and support vector machine (SVM)

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ABSTRACT

Image processing and artificial intelligence (AI) techniques have been applied to analyze, evaluate and classify mulberry fruit according to their ripeness (unripe, ripe, and overripe). A total of 577 mulberries were graded by an expert and the images were captured by an imaging system. Then, the geometrical properties, color, and texture characteristics of each segmented mulberry was extracted using two feature reduction methods: Correlation-based Feature Selection subset (CFS) and Consistency subset (CONS). Artificial Neural Networks (ANN) and Support Vector Machine (SVM) were applied to classify mulberry fruit. ANN classification with the CFS subset feature extraction method resulted in the accuracy of 100 %, 100 %, and 99.1 % and the least mean square error (MSE) values of 9.2×10^{-10}, 3.0×10^{-6}, and 2.9×10^{-3} for training, validation, and test sets, respectively. The ANN structure with the CONS subset feature extraction method resulted in the acceptable model with the accuracy of 100 %, 98.9 %, and 98.3 % and calculated MSE values of 4.9×10^{-9}, 3.0×10^{-3}, and 3.1×10^{-3} for training, validation, and test sets, respectively. In general, the machine vision system combined with the ANN and SVM algorithms successfully classified mulberries based on maturity. Finally, the ANN model with four features (R, B, b*, and Cr) selected through the CONS subset method with the least number of inputs and acceptable high classification accuracy with low MSE value was proposed as the proper model for online applications.

1. Introduction

Mulberry fruit are usually harvested manually by shaking the tree branches or mechanically using a shaker, resulting in a mixture of unripe, ripe, and overripe fruit. Different stages of ripeness result in low marketability and increased product waste. An essential post-harvest operation is the agricultural product classification based on their external features.

Machine vision, combined with artificial intelligence, has attracted a great deal of attention for use in agricultural and food industry applications, especially for recognition and classification of quality factors (Bargoti and Underwood, 2017; Teimouri et al., 2018; Azarmdel et al., 2019; Khojastehnazhand et al., 2019; Taheri-Garavand et al., 2019). Image processing and machine vision provides us with new knowledge that can be applied to control the quality of food and agricultural products with high accuracy (Nogales-Bueno et al.; 2014; Mohammadi et al.; 2015; Fashi et al., 2019).

Liming and Yanchao (2010) found that size detection error using image processing for strawberries was less than 5 %, color classification accuracy was 89 %, and shape classification accuracy was higher than 90 %. Mohammadi et al. (2015) found differences among different levels of ripeness of persimmons for most of the visual characteristics, including the R, G, and B channels, and the greyscale, and that the Quadratic Differential Analysis (QDA) was able to grade the fruit with 90 % overall accuracy. Wan et al. (2018) used computer vision and the backpropagation neural network method to detect three ripeness levels of tomato. Khojastehnezhad et al. (2019) used image processing to detect differences of apricots with respect to different features such as G, grayscale, L* and b*. In addition, the LDA and QDA techniques were able to classify apricots with 90.4 % and 92.3 % accuracies, respectively.

In addition to classification, the SVM algorithm can be used for both estimation and classification. Applications of the SVM algorithm can include pattern recognition, image processing and medical use, (Nouri

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et al., 2019; Ebrahimi et al., 2017; Peng et al., 2018). SVM has been applied for pattern and texture recognition, harvest time estimation, and product quality determination. Yu et al. (2014) proposed a new method for detecting fresh jujube cracks and maturity stages using ultra-high-resolution imaging in combination with image processing. They used differentiation models to identify mature and cracked fruit. Mizushima and Lu (2013) developed an automatically adjustable algorithm that graded the apple fruit based on its apparent color using the SVM and the Otsu thresholding method. In another study, Sun et al. (2014) determined the position and type of whiteness in rice grains with high accuracy by combining the SVM and image processing methods. A great deal of research has been conducted on the grading of agricultural products such as mango (Vélez-Rivera et al., 2014), apple (Jarolmasjed et al., 2016), potato (Al Riza et al., 2017), olive (Beyaz et al., 2017), banana (Larada et al., 2018), date (Nasiri et al., 2019), carrot (Jahanbakhshi and Kheiralipour, 2019), and pomegranate (Fashi et al., 2019) using machine vision.

The present study investigates the capability of the image processing method and classification techniques in grading mulberries based on their apparent characteristics. In this research, the emphasis is on the applicability of machine vision as a quick, inexpensive and accurate tool for grading mulberries in accordance with their level of ripeness. Thus, image processing algorithms and classification methods (ANN and SVM) were applied to extract mulberry features to classify the fruit in six classes with three ripeness levels (unripe, ripe, and overripe). Finally, a proposed grading system is presented, indicating the feasibility of applying machine vision and artificial intelligence for mulberry fruit classification.

2. Materials and methods

In this study, white (*Morus alba)* and red (*Morus rubra*) mulberry types were freshly harvested from an orchard in Tabriz - Iran. First, an expert graded samples of each type into three groups of unripe, ripe, and overripe fruit and the images of all six classes were captured. In total, 106 unripe, 92 ripe, and 100 overripe fruit were in the set of white mulberry, and 80 unripe, 109 ripe, and 90 overripe fruit were in the group of red mulberry (Figs. 1 and 2). In this study, the image acquisition system had a lighting case that consisted of 432 white and yellow LED lights installed inside the case. A Casio EX-H20 G 14.1 M P CCD digital camera was used to capture the images.

Modern devices for grading the agricultural products capture groups of samples in a single frame. By capturing each sample in one snapshot, the time spent for capturing and analyzing the mulberry samples will increase, and the capacity of the sorting system will decrease. Therefore to introduce the image processing algorithm for an online grading system, it is necessary to capture the samples all at once. Therefore, one image was obtained from all the samples of each mulberry class.

2.1. Selection of the appropriate color channel and mulberry segmentation

One of the most common methods for image segmentation is to apply the Otsu (1979) thresholding method. This algorithm converts the initial image into two classes (foreground and background) by minimizing the intra-class variances. By comparing the pixel intensities and the corresponding number of the pixels, the minimum value of the histogram is considered as the segmentation criteria. Note that, as shown in Fig. 3, if each mulberry type (white or red) was considered separately, any of the R, G, or B color channels could be used for image segmentation. However, the classification operation for mulberries with different levels of ripeness (unripe, ripe, and overripe) was performed for both types. Besides, to ensure the capability of the Otsu method to segment the fruit, the proposed algorithm was applied to classify a combination of samples with different levels of ripeness. Images of R, G, and B color channels for separated (left and right with smaller size) and combined white and red fruit (center with bigger size) are shown in Fig. 3.

As shown in Fig. 3, to segment each single mulberry class, any of the R, G, and B channels can be used, but in the combined samples with different levels of ripeness, an acceptable segmentation cannot be achieved in R and G color channels. Therefore, among the studied color channels, channel B showed the best segmentation for all samples. In the next step, any possible holes in the images were filled. In addition, some small objects rather than the fruit were observed beside the samples (non-target objects), which were also removed to avoid any disturbance in the feature extraction process.

2.2. Separating the stem ends from the mulberries

To accurately classify the mulberries and reduce the errors in the classification process, it is necessary to segment the mulberry area without any stems. Thus, after the initial segmentation, an erosive operator was used to separate the stems from the fruit (Fig. 4). This is performed by considering a structural element indicating the number of pixels in the neighborhood to filter the image by eroding the edges of the objects (Solomon and Breckon, 2011). Afterward, the image was dilated to return the mulberry area to the initial size with the least possible erosion.

2.3. Extraction of geometrical, color and texture features of mulberries

After the initial segmentation process, each fruit in the image was isolated from the initial image to extract the related features. Fig. 5 illustrates the smallest rectangular banding boxes around the samples in which the inside area is used to separate the color samples for the subsequent process.

As it is necessary to extract mulberry features apart from the rest of the image, the color image is overlaid on the binary image. Therefore, the coordinates of the mulberry samples in the primary color image, which match the position of "1s" (white area) of the binary image, are conserved, and the rest is converted to “0” (black area). Finally, the geometrical, texture, and color characteristics are determined for each mulberry sample. Features extracted from the mulberries are reported in Table 1.

2.4. Feature selection

CFS subset (CFS) and Consistency subset (CONS) methods were used to extract the most relevant features from the images. Using the CFS method, 14 top features were selected based on the ‘Best First’ criterion. Four top features were also selected using the CONS method. Initial and selected features based on both methods are presented in Table 1. WEKA software was used to reduce the input data dimension.

2.5. Classification algorithms

2.5.1. Artificial Neural Network (ANN)

Neural networks are simple modeling of real nervous systems. The scope of application of these networks is so broad that it includes applications from classification to those such as estimation and detection. An ANN is a set of simple structures (neurons, also called nodes) that together describe a final complex system. A set of nodes and connections between them are the two main components of a network. Each node is a network computing unit that can perform simple or complex processing on the inputs. Unlike mathematical models, neural network models can detect the relationships among parameters without extracting those relationships, and this has made the neural networks a powerful tool in classification and modeling (Jahanbakhshi and Salehi, 2019; Utai et al., 2019). To run the network, 65 % of all the input data were considered as training data, 15 % as validation, and 20 % as test data. We applied MATLAB 2017 software to process the images and run the artificial intelligent models.

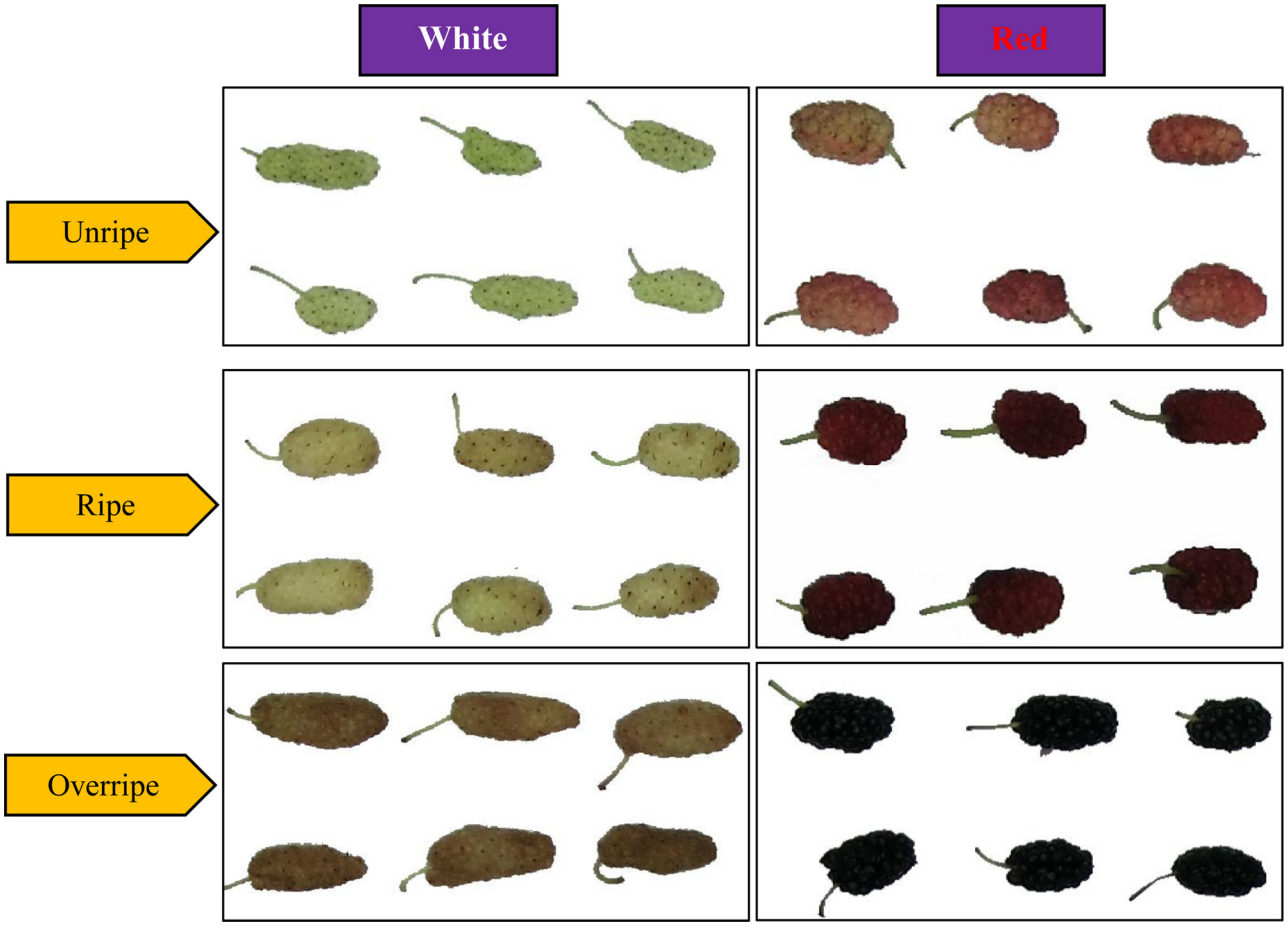


Fig. 1. Six different categories of two mulberry types (white and red).

In the next step, conventional hyperbolic tangent sigmoid (tansig) and logarithm sigmoid (logsig) transfer functions were used (Marti et al., 2013) to activate the weighted outputs. The following are the equations related to the tangent sigmoid and the logarithm sigmoid activation functions.

$$f(x) = \frac{2}{1 + e^{-2x}} - 1 \tag{1}$$

$$f(x) = \frac{1}{1 + e^{-x}} \tag{2}$$

In the hidden layer, a set of 24 neurons ranging from 2 to 25 was tried. To show the statistical reliability and develop an accurate model

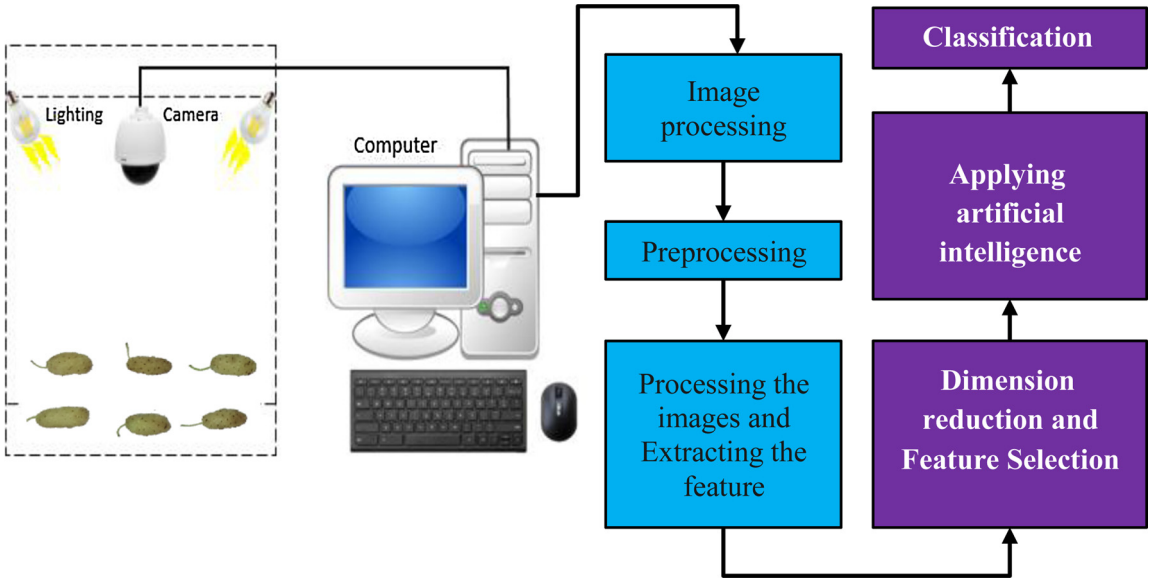


Fig. 2. Imaging system and mulberry classification.

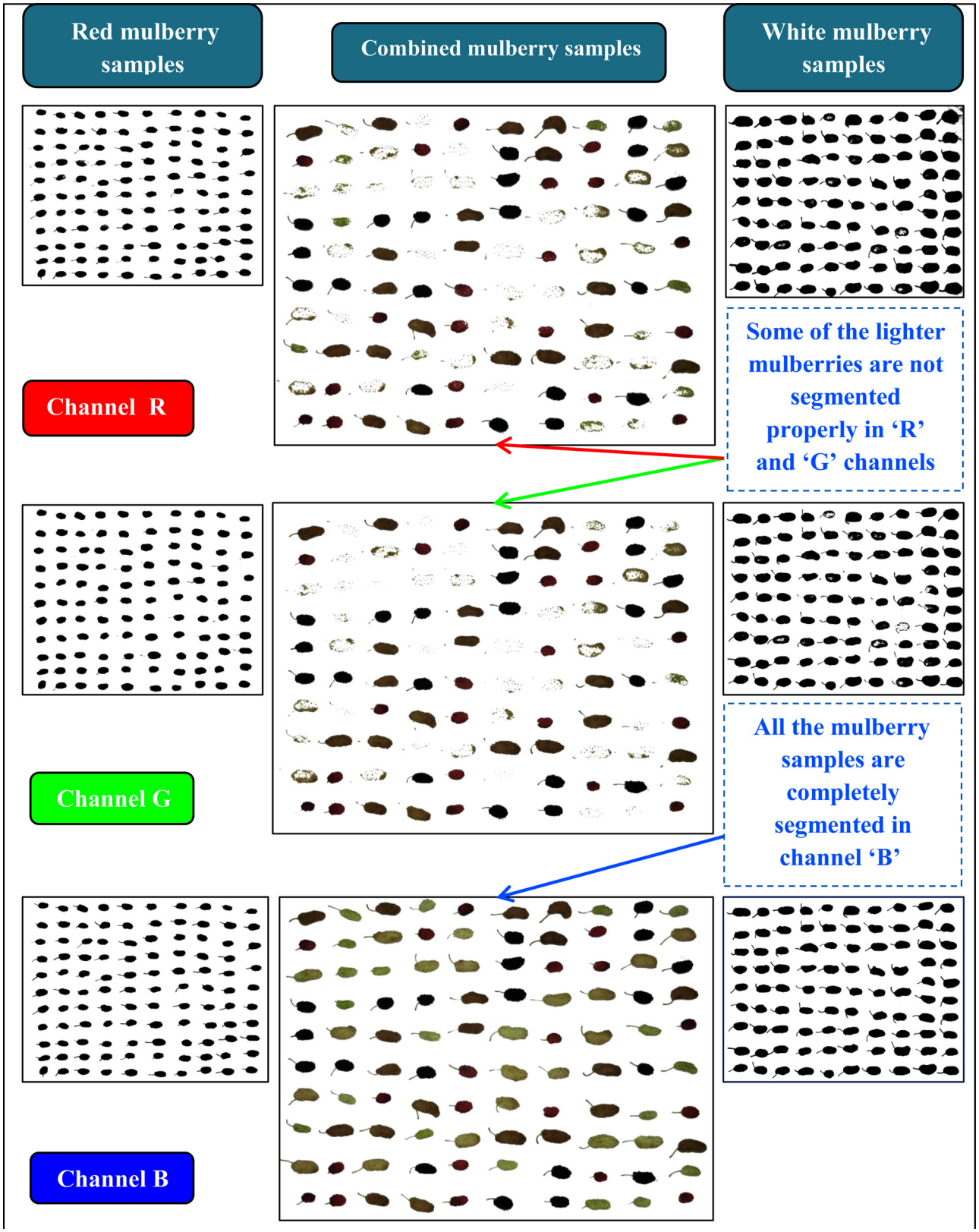


Fig. 3. Segmenting mulberry samples using the Otsu method in R, G and B color channels for red mulberries (left), white mulberries (right) and combined types (center).

on the selected structure of the ANN, researchers have tested different numbers of replications. Each structure was trained 50 times and the best value was recorded (Teimouri et al., 2016). The ANN structure and applied models are presented in Fig. 6.

Based on the two training algorithms ('trainlm' and 'traingdx'), two input data sets (CFS and CONS), and four different conditions resulted from the combination of two activation functions (Tansig and Logsig) in hidden and output layers, a total of 16 running structures were created.

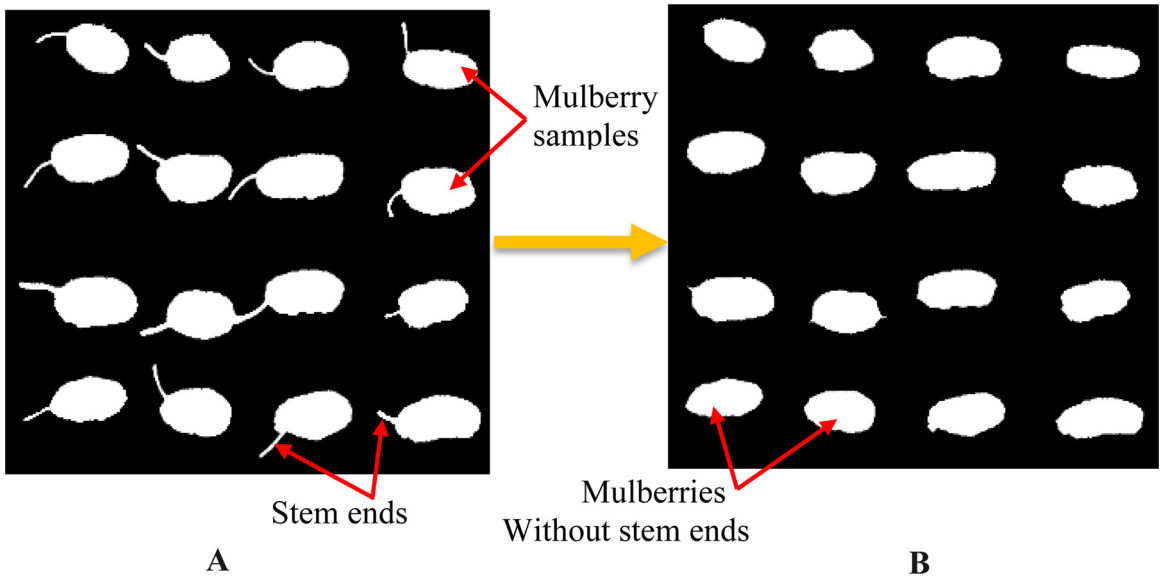


Fig. 4. Omitting the mulberry stem ends using edge erosion and dilation algorithm: A) mulberry samples with stem ends, and B) final binary image without stem ends.

By taking into account the number of 24 neurons in the hidden layer and repeating the same structure for 50 times, a total of 1,200 (24 × 50) iterations were conducted for each of the 16 structures. A total of 19,200 (1200 × 16) runs were conducted to judge and select the best ANN structure suited for mulberry classification based on the least mean MSE value. The best ANN selection process is presented in Fig. 7.

2.5.2. Support Vector Machine (SVM)

The SVM algorithm is one of the most accurate and robust algorithms in data mining and can be used to classify linear and nonlinear data. The SVM algorithm was first proposed by Cortes and Vapnik (1995) to classify data that were linearly separable and later generalized to nonlinear states. By using this algorithm, data are categorized into two phases of train and test sets, and to validate the data, cross-validation techniques such as k-fold, holdout, or leave-n-out training methods are applied. In order to find the optimal value of the penalty parameter (C) and the kernel parameter (σ), the holdout cross-validation was applied to perform the classification with the highest accuracy. For this purpose, 20 % of the whole data was considered as test data each time, and the rest was taken as training data.

SVM is a binary classification model. This algorithm converts the training data space into a higher dimension through nonlinear mapping and then looks for a hyperplane in this new dimension to separate samples of one class from the other classes. This algorithm uses concepts such as support vectors and margins to find this hyperplane. A

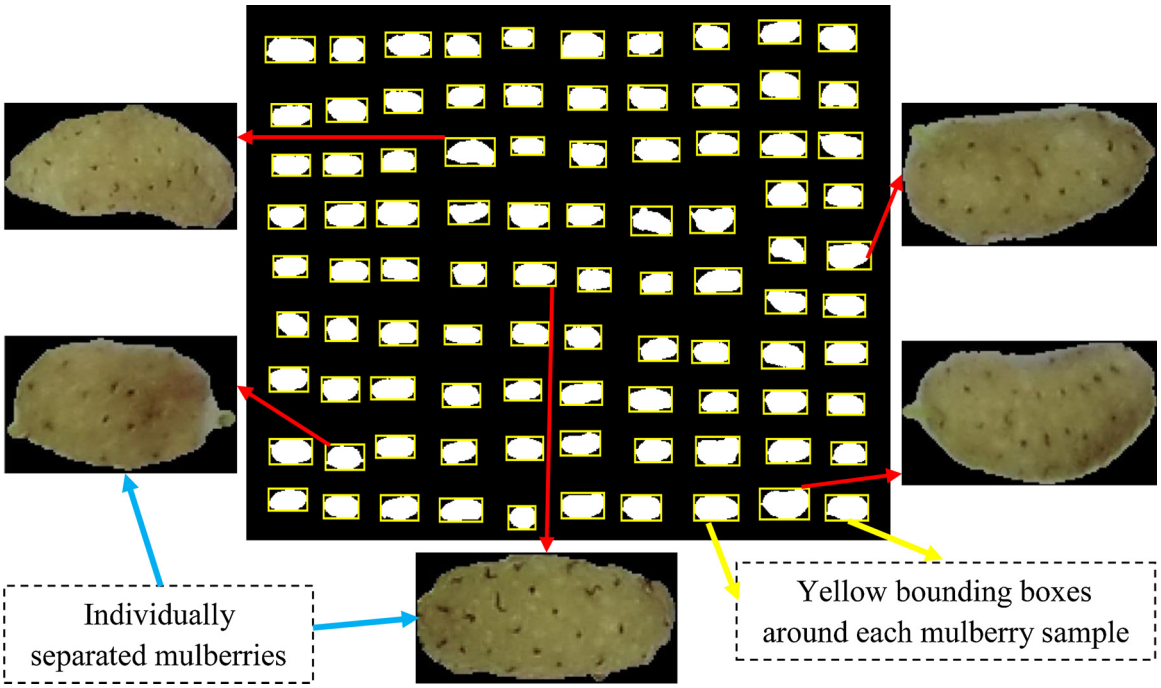


Fig. 5. Separating the mulberries in the initial image as the individual samples by matching the white pixels “1s” of the binary image with the initial color image based on the rectangular bounding box.

Table 1
Extracted and selected features using CFS subset (CFS) and Consistency subset (CONS) methods.

Total extracted features			Selected features based on evaluation method		
			CFS subset (CFS)		
Color features (Mean)	Geometric Features	Texture features	Color features (Mean)	Geometric Features	Texture features
R-(RGB)	Area	Standard deviation	G-(RGB)	Perimeter	Contrast
G-(RGB)	Perimeter	Entropy	B-(RGB)	Equivalent Diameter	Homogeneity
B-(RGB)	Eccentricity	Contrast	Gray	Area/Perimeter	
Gray	Extent	Homogeneity	L-(La*b*)		
L*-(La*b*)	Orientation	Correlation	a*-(La*b*)		
a*-(La*b*)	Equivalent Diameter	Energy	b*-(La*b*)		
b*-(La*b*)	Max axis/Min axis		M- (CMY)		
H-(HSV)	Area/Perimeter		Cb-(YCbCr)		
S-(HSV)	Roundness		Cr-(YCbCr)		
V-(HSV)					
C- (CMY)					
M- (CMY)			Consistency subset (CONS)		
Y-(CMY)					
Y-(YIQ)			R-(RGB)		
I-(YIQ)			B-(RGB)		
Q-(YIQ)			b*-(L*a*b*)		
Cb-(YCbCr)			Cr-(YCbCr)		
Cr-(YCbCr)					

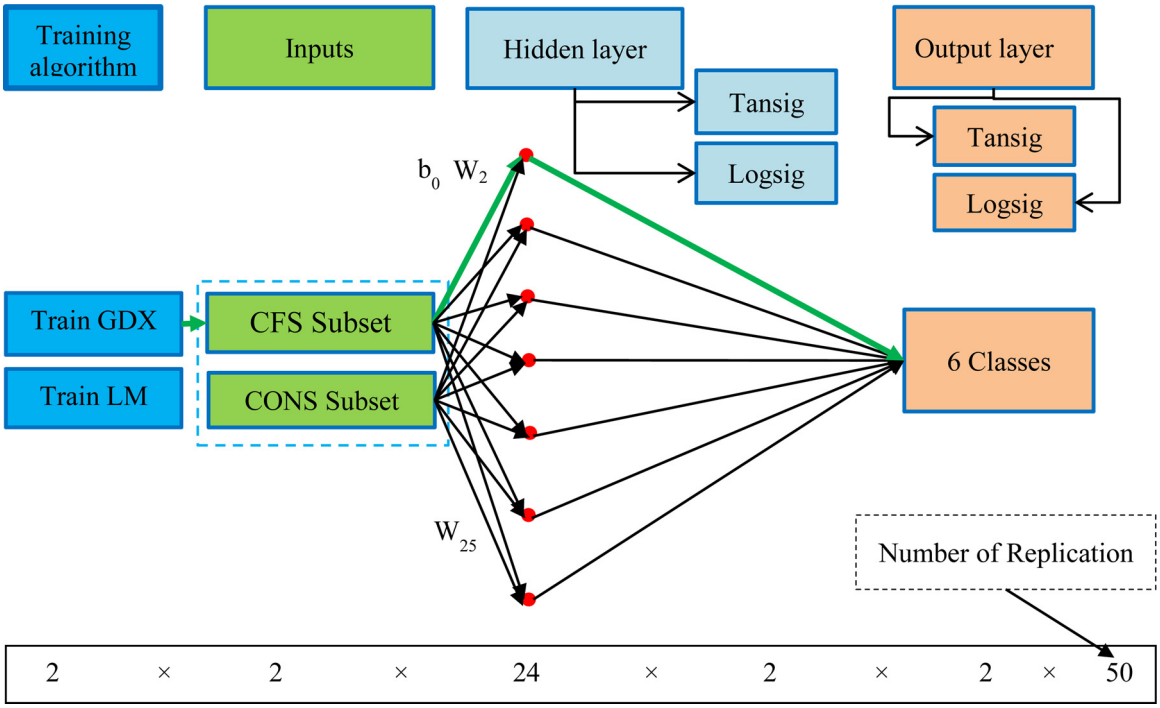


Fig. 6. The ANN structure with the number of the input features, the number of the nodes in the hidden layer, the activation functions and the training algorithms.

sequence of binary clusters was used to categorize several classes. Assuming that each of the six classes used in this study are linearly separable in pairs, the SVM binary class was trained for 15 times as a permutation of T_6^2 by the SVM algorithm, and the final result was determined based on the sum of the results in each voting (Wu et al., 2018; Mohammadi and Minaei, 2019).

To classify nonlinearly separable data, the data are mapped onto space with larger dimensions (feature space). Such algorithms require complex computation, which results in slow functioning. Therefore, kernel functions are used to separate nonlinear data linearly so that based on these transformations, an optimal boundary could be drawn for separation, which is not necessarily a straight line. The advantage of these kernels is creating connections between data points and simplifying the complex computations.

Some of the kernels used in SVM are radian basis or Gaussian functions, polynomial, or linear functions. Therefore, the MSE values were calculated along with the classification accuracy to find the optimal 'C' and 'σ' values. Wu et al (2018) used RBF kernels in their study with (C, σ) = {10, 20, …, 100} × {1, 1.5, 2} and polynomials with (C, p) = {1, 2, …, 100} × {2, 3}. Since in SVM it is important to find the appropriate values of 'C' and 'σ' to define the best network structure with the highest classification rate and the lowest error rate, the intervals with smaller steps with (C, σ) = {1, 2, …, 100} × {0.25, 0.5, …, 2.5} and polynomials with (C, p) = {1, 2, …, 100} × {2, 3} were tried in this study. The linear kernel was also applied to evaluate the classification accuracy and compare it with the results of RBF and polynomial kernels. In general, an algorithm with 10400 iterations was implemented to extract the optimal 'C' and 'σ' values using input features of both CFS and CONS methods.

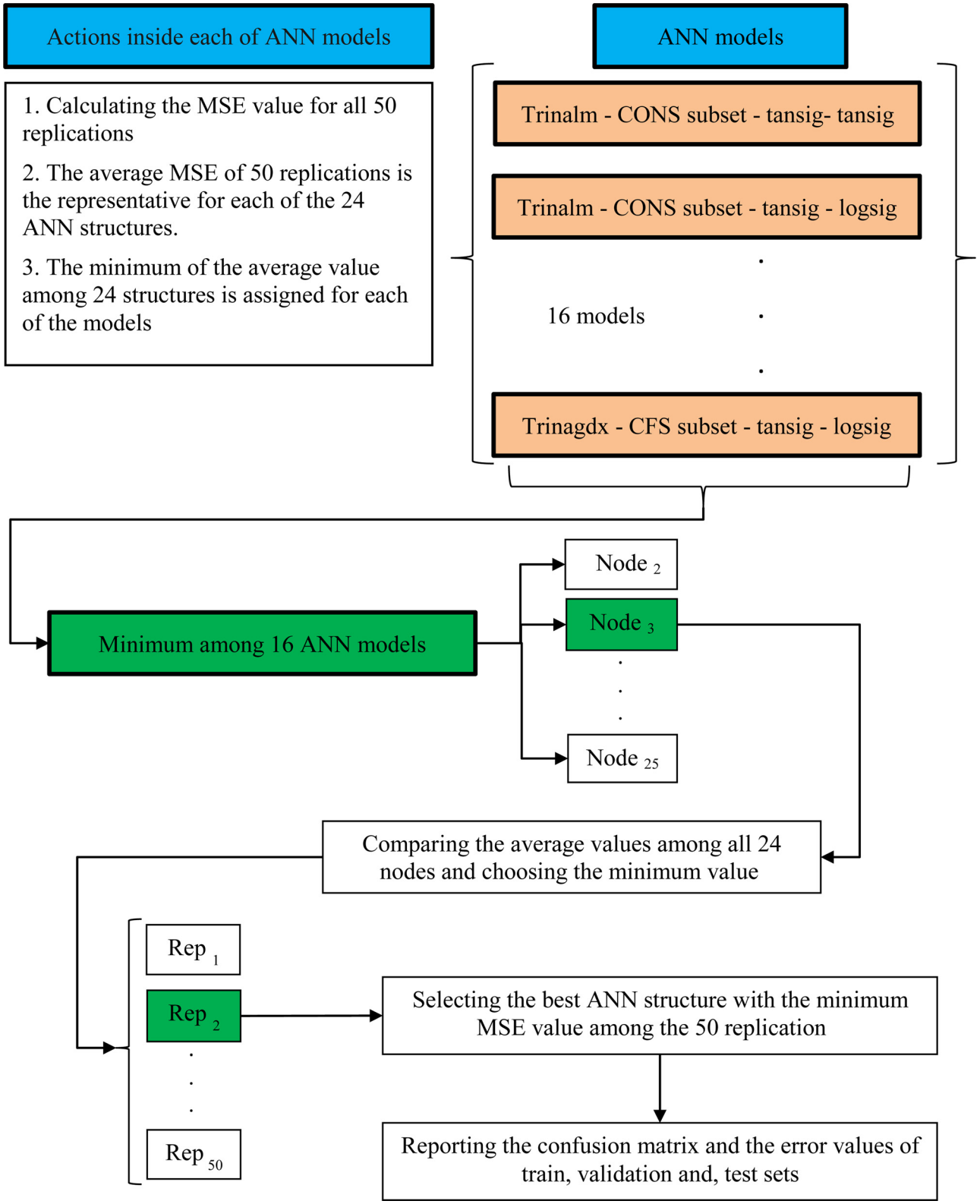


Fig. 7. The process of selecting the best ANN structure for mulberry classification.

2.6. Comparing the results

After the network structure was determined, the data were imported to the neural networks. To normalize the input data in the [-1 − 1] and [0 – 1] intervals, depending on the activation function (tansig and logsig) of the network output, equations (3) and (4) were used respectively (Marti et al., 2013). To train the network using the SVM algorithm, the input data was normalized between [0 – 1].

$$Y_N = \frac{Y_i - Y_{\min}}{Y_{\max} - Y_{\min}} \quad (3)$$

$$Y_N = 2 \times \left(\frac{Y_i - Y_{\min}}{Y_{\max} - Y_{\min}}\right) - 1 \quad (4)$$

In these relations, Y_i represented the input data, Y_{min} was the minimum input data, Y_{max} was the maximum input data and Y_N showed the normalized data in the given interval.

To compare the performance of the ANN and SVM algorithms, the Mean Square Error (MSE) was calculated as described in equation (5) (Emamgholizadeh et al., 2015).

$$MSE = \frac{\sum_{i=1}^{n} (o_i - p_i)^2}{n} \quad (5)$$

In this relation, n, o_i, and p_i were the number of data, the values observed, and the values predicted, respectively.

In order to evaluate the accuracy of the given algorithm, three statistical parameters, namely sensitivity (SEN), specificity (SPE) and accuracy (ACC), were used. Each parameter was calculated as described in equations (6), (7), and (8), respectively (Omid, 2011).

$$Sensitivity = \frac{TP}{(TP + FN)} \quad (6)$$

$$Specificity = \frac{TN}{(TN + FP)} \quad (7)$$

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FN + FP)} \quad (8)$$

In these relations, True Positives (TP) are the number of samples that are correctly classified in group A. True Negatives (TN) are the number of samples belonging to other groups. False Negatives (FN) are the number of samples belonging to group A which are incorrectly classified in other groups. False Positives (FP) are the number of samples of other groups that are incorrectly classified as group A.

3. Results and discussion

3.1. ANN results

Output data for fruit were divided into three categories (training, validation, and test data). To evaluate the performance of activation functions in fruit classification, two activation functions were tried in the hidden and output layers to train the neural network: tangent sigmoid (TAN) and logarithm sigmoid (LOG).

To evaluate the network structure and to ensure its reliability, 50 replications were assigned to each node in the network structure. The mean MSE values of training, validation, and test sets were determined for each of the replicates. Then, the best network structure was selected based on the lowest total mean MSE value among the tried number of the nodes (2-25 nodes). By considering the combination of the LOG and TAN activation functions in both hidden and output layers, the results for the four modes were obtained. As mentioned above, two sets of different inputs were introduced to the network based on the CFS and CONS feature selection methods. Four graphs containing four lines each of which represents one of the network structures are shown in Fig. 8.

Figs. 8A and 8C are related to the network training based on the Levenberg-Marquardt (LM) algorithm. Figs. 8B and 8D belong to the gradient descend training (GDX) algorithm. The analyses show higher MSE values in the LOG-TAN (logarithm sigmoid activation function in the hidden layer and tangent sigmoid activation function in the output layer) network structure. Therefore, using the TAN function in the output layer results in higher error values compared to the other modes. The results also show that using the LM algorithm (Figs. 8A and 8C), the MSE values face many fluctuations (red line) without a constant trend by increasing the number of neurons in the hidden layer. The lowest error values in all four figures belong to the TAN-LOG and LOG-LOG structures (blue and brown lines) (Fig. 8).

After selecting the appropriate network structure from all the cases studied, the best structure with the least error value was selected as the optimal structure. Fig. 9 shows the minimum error values among all 16 models. The values are presented for 16 different models, resulting from the combination of activation functions (TAN and LOG), network learning algorithms (LM and GDX), and input selection methods (CONS and CFS).

In all network structures, the error value calculated by using the CFS feature extraction method is lower than the value obtained through the CONS method (Fig. 9), which may be due to a large number of the features in the CFS method. The higher the number of network inputs, the higher the computational burden on the network. If the final classification function (optimal structure) is used in any online processing systems (machines used to process mulberries), the large number of inputs will increase the computation time. The best network structure based on the lowest MSE using CFS and CONS methods was the LM-T-L and LM-L-L structures with values of 1.3×10^{-3} and 2.1×10^{-3}, respectively. It should be mentioned that the difference in error values between these two cases is close to each other, which shows the selected four features with the CONS method can be considered as an acceptable model compared to 14 features selected with the CFS method. This is discussed in the following, and the results are presented in detail.

Table 2 presents the confusion matrix for the best ANN structure using the CFS method. The results of the training and validation data show that all samples are correctly classified in their groups with 100 % accuracy. The sensitivity of the BR1 resulted in 94.44 % for the test set. On the other hand, the specificity of BR2 resulted in 98.85 %, which shows that misclassified samples of BR3 fall in this group. The total accuracy of the network for the test set resulted in 99.13 %. The MSE values of 9.2×10^{-10}, 3.0×10^{-6}, and 2.9×10^{-3} are calculated for the training, validation, and test sets, respectively.

The neural network results using the CONS subset feature extraction method are reported in Table 3. Using this subset for classification showed the sensitivity, specificity, and accuracy of 100 % for the training dataset. The sensitivity value of 94.44% was calculated for BW2 in the validation dataset. On the other hand, the specificity value of the BR1 resulted in 98.59 %, indicating that the misclassified samples were fallen in this group.

The results of the test set showed the sensitivity values of 95.24% and 92.31% for BW3 and BR1, respectively. On the other hand the specificity value of 98.96 % and 98.94 % were calculated for the BW2 and BW3, respectively. Besides, the total accuracy of the test set resulted in 98.26 %. The MSE values of 4.9×10^{-9}, 3.0×10^{-3}, and 3.1×10^{-3} were obtained for the training, validation, and test sets, respectively.

Results show that the average of MSE for 50 replications using the CFS feature selection method was lower than that of the CONS method (Fig. 9). Still, the results indicate a more accurate classification of different groups of mulberries through the CONS method (Table 3) with lower error values. Although the average MSE value of CFS based selected features is lower than that of CONS, the related structure trained with the CONS subset features showed lower MSE value as we have reported the lowest MSE error of each structure among 50 tried replications. Although, the selected representative structure of the LM-T-L CFS subset resulted in lower MSE value compared to the LM-L-L of CONS subset, considering a network structure with lower input features has priority over a large number of the features.

3.2. SVM results

Table 4 presents the results of the classification of six groups of fruit by SVM algorithm using two different feature selection methods (CFS subset and Consistency subset) as well as the MSE values and the classification values of training and test sets using RBF, polynomial, and linear kernel functions. The optimal values of 'C' and 'σ' for using different modes are also presented in Table 4. The least error values by

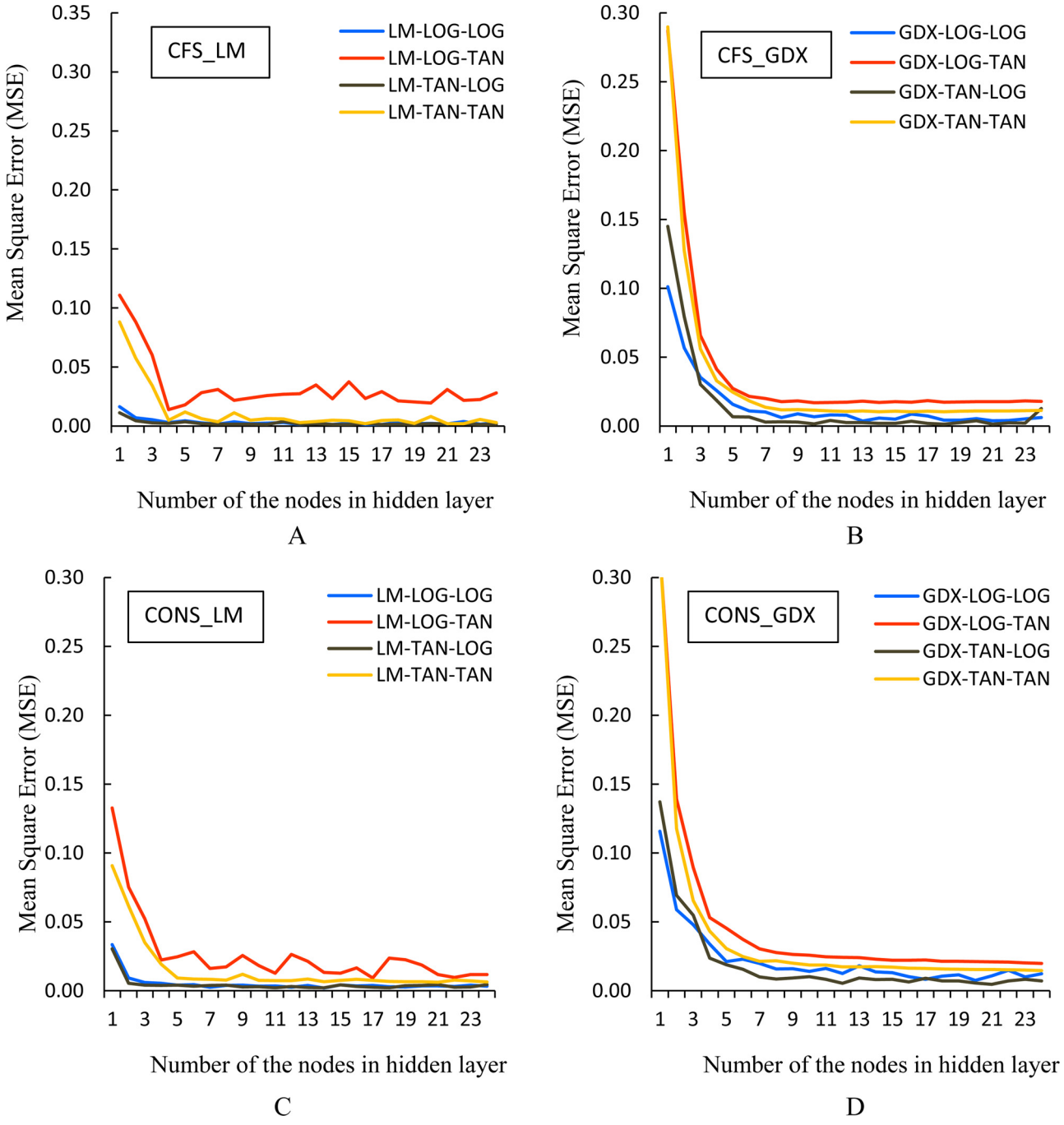


Fig. 8. Results of mean square error (MSE) for different ANN topologies using the Levenberg Marquardt (LM) and gradient descend training (GDX) algorithms in both (CFS) and (CONS) subsets. Four different network structures based on the activation function of the nodes (logarithm sigmoid (LOG) or tangent sigmoid (TAN)) in both hidden and output layers were tried.

using the CFS subset and CONS subset belong to the RBF and polynomial kernels, respectively. It should be noted that the polynomial functions with p = 2 are the same functions known as quadratic functions.

The SVM confusion matrix for the CFS and CONS subsets is given in Table 5. In addition to the low error rate of the CFS subset method with an RBF kernel function, the sorting accuracy is also higher compared to the CONS subset with a polynomial function. In fact, among all the combinations under study, the best SVM network structure for sorting fruit is the CFS subset with an RBF kernel (Table 4). The MSE values of the CFS and CONS subsets show the lowest MSE values of 9×10^{-3} and 9×10^{-3} for training and test sets, respectively. The MSE values of the CONS subset using the polynomial kernel are 1.7×10^{-2} and 1.8×10^{-2} for the training and test sets, respectively. Among both of the subsets, using the RBF kernel function with the CFS subset feature extraction method showed the lowest MSE. The classification accuracies concerning the CFS subset mode were 99.78 % and 99.12 % for training and test sets, respectively. These values for polynomial function using the CONS subset were 99.57 % and 98.25 %.

The results of the current study show that the automated vision system can successfully classify fruit into three maturity levels based on the color, shape, and texture using ANN and SVM classifiers. Among them, the best classification resulted in the ANN classifier using CONS subset, with the least number of inputs, low error value, and acceptable

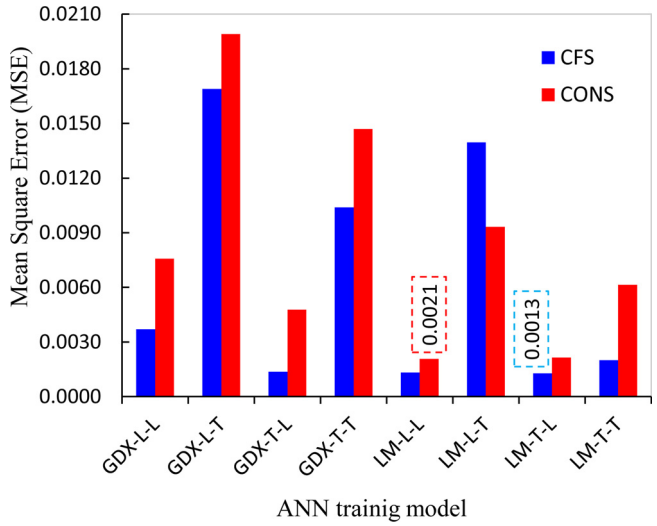


Fig. 9. Total Mean Square Error (MSE) values of 24 network structures with 50 replications per node. Each term in the horizontal axis represents the applied functions: applied training algorithm (LM or GDX) - activation function of the hidden layer (TAN or LOG) - activation function of the output layer (TAN or LOG).

Table 2
Confusion matrix of the Artificial Neural Networks (ANN) using the CFS subset (CFS) feature selection method with 20 neurons in the hidden layer (14-20-6).

	Train			Validation			Test		
	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)
BW1	100	100	**100**	100	100	**100**	100	100	**99.13**
BW2	100	100		100	100		100	100	
BW3	100	100		100	100		100	100	
BR1	100	100		100	100		94.44	100	
BR2	100	100		100	100		100	98.85	
BR3	100	100		100	100		100	100	
MSE	**9.2×10^{-10}**			**3.0×10^{-6}**			**2.9×10^{-3}**		

Note: Mean Square Error (MSE); Sensitivity (SEN); Specificity (SPE); Accuracy (ACC); three classes of white mulberry ($BW_{1\text{-}3}$); three classes of red mulberry ($BR_{1\text{-}3}$).

Table 3
Confusion matrix of Artificial Neural Networks (ANN) using the Consistency subset (CONS) feature selection method with 15 neurons in the hidden layer (4-15-6).

	Train			Validation			Test		
	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)
BW1	100	100	**100**	100	98.59	**98.85**	100	100	**98.26**
BW2	100	100		94.44	100		100	98.96	
BW3	100	100		100	100		95.24	98.94	
BR1	100	100		100	100		92.31	100	
BR2	100	100		100	100		100	100	
BR3	100	100		100	100		100	100	
MSE	**4.9×10^{-9}**			**3.0×10^{-3}**			**3.1×10^{-3}**		

Note: Mean Square Error (MSE); Sensitivity (SEN); Specificity (SPE); Accuracy (ACC); three classes of white mulberry ($BW_{1\text{-}3}$); three classes of red mulberry ($BR_{1\text{-}3}$).

high classification accuracy. Therefore, the proposed algorithm could objectively, rapidly, and effectively classify mulberry fruit by using a non-destructive vision technique.

4. A proposed plan for mulberry classification

Fig. 10 shows the proposed plan of a sorting system for mulberry classification. Fruit move into parallel grooves by the initial conveyor. Also, the inclined position of the vibrating plate guides the fruit downwards. The movement of the second conveyor belt carries the fruit over the holes in the belt as they pass on in front of a camera installed along with the device. After detecting the mulberry type through the machine vision and processing section, the machine orders the specific mulberry shooter by compressed air (pneumatic) to shoot the mulberry towards the proper quality spot at the right time. Among the six mulberry classes, the sixth one falls into its box from the end of the conveyor belt without shooting. This group can be used for one of the sound classes like BW2 or BR2. The other sound class can be guided from the shortest channel with the minimum amount of air pressure. The holes (seats) in the second conveyor belt are made so that samples are placed in stable positions. Moreover, there are holes designed to let the airflow contact over mulberry samples. The airflow rate is adjusted to avoid any bruise or crush on mulberry fruit.

5. Conclusion

In this study, white and red mulberry fruit were classified according to maturity stage using image processing and artificial intelligence classification algorithms. First, mulberry image segmentation was performed using the RGB color space. Among the tested color channels, the channel 'B' was selected as the best channel to classify fruit into three unripe, ripe, and overripe categories. In the next step, color, geometric, and texture features were extracted with two feature selection methods, namely CFS and CONS. After the image processing step, feature extraction, and dimension reduction, ANN and SVM were applied to classify each fruit as one of the six possible classes. Comparing the performance of the two methods (ANN and SVM), the ANN showed a significant advantage over the SVM for the mulberry classification. The best classification performance was obtained by using the CFS subset feature extraction method (14 selected features) with ANN. Although the CFS method showed the least error value with the highest classification rate, the final model was proposed based on both error value and number of inputs. Since this research aims to develop a combined machine vision and artificial intelligence algorithm for an online application, therefore, it is proposed to use the CONS feature subset to reduce the dimension of the initial data and introduce them to the ANN with the LM-L-L (Levenberg-Marquardt-Logsig-Logsig) structure to obtain acceptable classification with low MSE value and the least number of inputs. Thus, this structure is proposed to be used in a six-class mulberry sorting system: two types and three ripeness levels. According to the results of this study, machine vision can be used as an accurate, fast, and non-destructive method to classify and grade mulberry fruit, reduce product loss and increase its marketability.

Declaration of Competing Interest

The authors have declared no conflict of interest.

CRediT authorship contribution statement

Hossein Azarmdel: Methodology, Investigation, Software, Writing - original draft, Formal analysis. **Ahmad Jahanbakhshi:** Conceptualization, Methodology, Investigation, Writing - original draft, Data curation. **Seyed Saeid Mohtasebi:** Supervision, Data curation, Validation, Writing - review & editing. **Alfredo Rosado Muñoz:** Data curation, Validation, Writing - review & editing.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the

Table 4
Results of Support Vector Machine (SVM) for mulberry classification with two sets of selected features (CFS subset (CFS) and Consistency subset (CONS)) using three different kernel functions (RBF, polynomial, and linear).

Applied kernels	Number of iterations	MSE		Accuracy (%)		Poly order	Optimized parameters	
		Train	Test	Train	Test		C	σ
CFS subset (CFS)								
RBF	100 × 4 × 10	**0.009**	**0.009**	**99.78**	**99.12**	———	1	1.75
polynomial	100 × 4 × 2	0.000	0.114	100.00	98.25	2	1	———
Linear	100 × 4	0.009	0.035	99.78	99.12	1	5	———
Consistency subset (CONS)								
RBF	100 × 4 × 10	0.071	0.149	99.35	98.25	———	1	1
polynomial	100 × 4 × 2	**0.017**	**0.018**	**99.57**	**98.25**	2	2	———
Linear	100 × 4	0.073	0.079	99.57	99.12	1	9	———

Note: Mean Square Error (MSE); Radial Basis Function (RBF); penalty parameter (C); kernel parameter (σ).

Table 5
Confusion matrix of Support Vector Machine (SVM) using CFS subset (CFS) and consistency subset (CONS).

	CFS subset (CFS)						Consistency subset (CONS)					
	Train			Test			Train			Test		
Class	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)
BW1	100	100	**99.78**	100	100	**99.12**	100	100	**99.57**	100	100	**98.25**
BW2	100	100		94.74	100		100	100		90.00	100	
BW3	100	100		100	98.95		100	100		100	97.92	
BR1	100	99.73		100	100		98.86	99.73		100	100	
BR2	100	100		100	100		100	100		100	100	
BR3	98.46	100		100	100		98.44	99.75		100	100	
MSE	9.0×10^{-3}			9.0×10^{-3}			1.7×10^{-2}			1.8×10^{-2}		

Note: Mean Square Error (MSE); Sensitivity (SEN); Specificity (SPE); Accuracy (ACC); three classes of white mulberry ($BW_{1\text{-}3}$); three classes of red mulberry ($BR_{1\text{-}3}$).

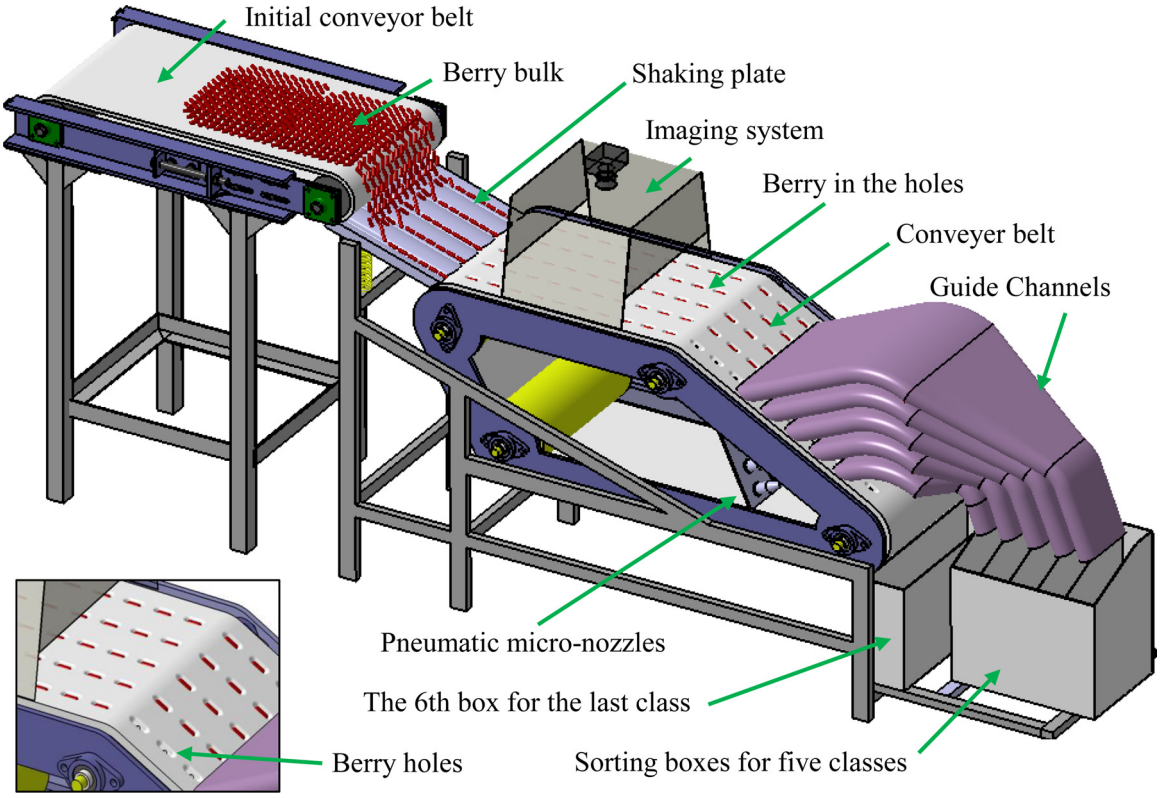


Fig. 10. The design of the proposed system for mulberry fruit classification.

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